INCH-POUND

MS24143W
24 Nov 1993
SUPERSEDING
MS24143V
31 March 1992

MILITARY SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 25 AMPERES, 3 PST (N.O.), TYPE I, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

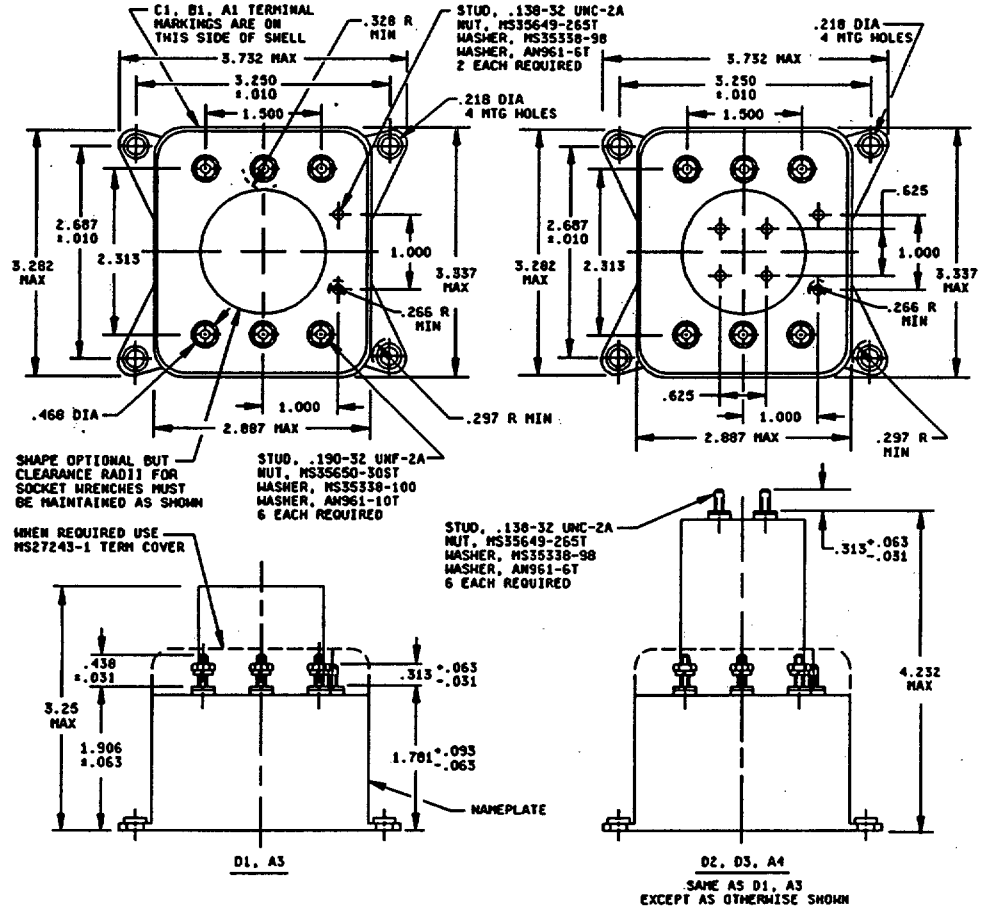


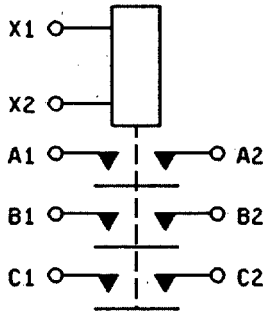

FIGURE 1. Dimensions and configurations.

denotes changes



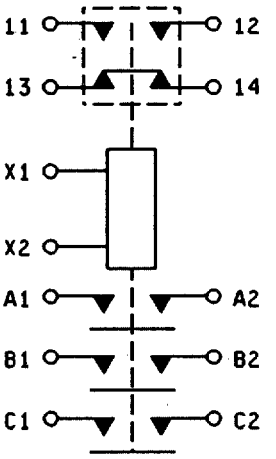
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CIRCUIT DIAGRAM A

APPLICABLE TO RELAYS
WITHOUT AUXILIARY CONTACTS




CIRCUIT DIAGRAM B

APPLICABLE TO RELAYS
WITH AUXILIARY CONTACTS

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .010 | 0.25 | .313 | 7.95 | 2.313 | 58.75 |
| .031 | 0.79 | .328 | 8.33 | 2.687 | 68.25 |
| .063 | 1.60 | .438 | 11.13 | 2.887 | 73.33 |
| .093 | 2.36 | .468 | 11.89 | 3.250 | 82.55 |
| .138 | 3.51 | .625 | 15.88 | 3.282 | 83.36 |
| .190 | 4.83 | 1.000 | 25.40 | 3.337 | 84.76 |
| .218 | 5.54 | 1.500 | 38.10 | 3.732 | 94.79 |
| .266 | 6.76 | 1.781 | 45.24 | 4.232 | 107.49 |
| .297 | 7.54 | 1.906 | 48.41 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Coil and auxiliary terminals may use additional washer for terminal seat.
5. This specification sheet takes precedence over procurement documents referenced herein.
6. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

ENVIRONMENTAL CHARACTERISTICS:

Temperature range 1/: -70°C to +125°C.
Maximum altitude rating: 80,000 ft.
Shock G-level: 25 g's.
Duration: 6-9 ms.
Max duration contact opening: 2 ms.
Vibration - Sinusoidal: See table I.
Vibration - Random: Not applicable.
High shock: Not applicable.
Acceleration: 15 g's.

ELECTRICAL CHARACTERISTICS (see tables II, III, and IV):

Insulation resistance, initial: 100 MΩ.
After life or environmental tests: 50 MΩ.
Dielectric strength (sea level): 2-5 s.

| | Initial 2/ | | After life tests 2/ | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac | 28 V dc | 115 V ac |
| Coil to case | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |
| Aux contacts | 1,250 V rms | 1,500 | 1,000 V rms | 1,125 |
| All other points | 1,250 V rms | 1,800 | 1,000 V rms | 1,350 |

Dielectric strength (altitude): 1 minute.

| | 28 V dc 2/ | 115 V ac 2/ |
|---|---|---|
| Coil to case | 500 V rms | 500 |
| Aux contacts | 500 V rms | 500 |
| All other points | 700 V rms | 700 |

Max contact drop initial: .150 V.
After life test: .175 V.
Overload current (NO): 200 A. 3/
Rupture current (NO): 250 A. 3/
Duty rating: Continuous.
RFI specification: MIL-STD-461.
(Applicable to coil circuits of ac operated relays.)

1/ Coils operated at either 50, 60, or 400 Hz require ambient temperature not to exceed +71°C maximum.
2/ For A3 and A4, coil terminals X1 and X2 must be shorted together for all dielectric testing between coil to case, coil to main or auxiliary contacts, and coil to all other points.
3/ For A3 and A4, rupture and overload current (NO) is 75 A.



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TABLE I. Vibration levels.

| Dash number | 5-10 Hz | 10-55 Hz | 55-250 Hz | 250-500 Hz | 500-1,500 Hz |
|---|---|---|---|---|---|
| D1 | .06 DA | .06 DA | 10 g's | 6 g's | 6 g's |
| D2 | | | | 5 g's | 4 g's |
| D3 | | | | 5 g's | 4 g's |
| A3, A4 | | | | 5 g's | 4 g's |

TABLE II. Operating characteristics.

| Dash number MS24143- | Coil data | | | | | | | | | | | Time - milliseconds max. | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Rated | | Max | | Max pick-up voltage | | | | Hold voltage 2/ 3/ | Drop out voltage 2/ 3/ | Operate 4/ | Release 5/ | Contact bounce 4/ | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | Coil | Volts 1/ | Frequency Hz | Res +15% -10 | Volts | Ampere | Normal 2/ | High temp test | Cont current test | | | | | NO | NC | NO | NC |
| D1 | X1,X2 | 28 | dc | 52 | 29 | 0.6 | 18 | 21 | 22.5 | 7 | 1.5 | 25 | 10 | 2 | --- | --- | --- |
| D2, D3 | X1,X2 | 28 | dc | 52 | 29 | 0.6 | 18 | 21 | 22.5 | 7 | 1.5 | 25 | 10 | 2 | --- | 4 | 4 |
| A3 | X1,X2 | 115 | 50/60/400 6/ | --- | 124 | .225 | 90 | 100 | 104 | 40 | 10 | 30 | 65 | 5 | --- | --- | --- |
| A4 | X1,X2 | 115 | 50/60/400 6/ | --- | 124 | .225 | 90 | 100 | 104 | 40 | 10 | 30 | 65 | 5 | --- | 6 | 6 |

1/ Caution: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ At 50/60 Hz, chattering may occur at or near dropout voltage when voltage is slowly decreased.
4/ With rated coil voltage.
5/ From rated coil voltage.
6/ Coils will operate on either 50, 60, or 400 Hz except ambient temperature must be derated to +71°C maximum.





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(W) TABLE III. Rated contact load (amperes per pole) case grounded (applicable to -D1, -D2, -A3, and -A4).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1-phase | | | | 115/200 V ac, 3-phase 1/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main 2/ | | Aux | |
| | | NO | NC | NO | NC | 400 Hz | 50/60 Hz | 400 Hz | 50/60 Hz | 400 Hz | 50/60 Hz | 400 Hz | 50/60 Hz |
| Resistive | 50 | 25 | --- | 5 | 5 | 25 | 25 | 5 | --- | 25 | 15 | --- | --- |
| Inductive | 50 | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| Inductive | 10 3/ | 25 | --- | 5 | 5 | 25 | --- | 5 | --- | 25 | 12.5 | --- | --- |
| Motor | 50 | 25 | --- | --- | --- | 25 | 15 | --- | --- | 25 | 12.5 | --- | --- |
| Lamp | 50 | --- | --- | .75 | .75 | --- | --- | .75 | --- | --- | --- | --- | --- |
| Transfer load | | | | | | | | | | | | | |
| Mechanical life reduced current | 100 | 6.25 | --- | 1.25 | 1.25 | 6.25 | 6.25 | 1.25 | --- | 6.25 | 3.75 | --- | --- |
| Intermediate current | 50 | 5 | --- | Per spec | Per spec | --- | --- | --- | --- | --- | --- | --- | --- |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Derated contact loads at 50/60 Hz are applicable to A3 and A4 only.
3/ Derate cycles by 50 percent for A3 and A4 only.





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(W) TABLE IV. Rated contact load (amperes per pole) case grounded (applicable to -D3 only).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1-phase | | | | 115/200 V ac, 3-phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 12 | --- | 1.25 | 1.25 | 12 | --- | 1.25 | --- | 12 | --- | --- | --- | |
| Inductive | 100 | 12 | --- | .75 | .75 | 12 | --- | .75 | --- | 12 | --- | --- | --- | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 100 | 6 | --- | --- | --- | 6 | --- | --- | --- | 6 | --- | --- | --- | |
| Lamp | | | | | | | | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 100 | 3 | --- | .25 | .25 | 3 | --- | .25 | --- | 3 | --- | --- | --- | |
| Intermediate current | 100 | 1.2 | --- | Per spec | | 1.2 | --- | --- | --- | 1.2 | --- | --- | --- | |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.





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Part or Identifying Number (PIN): MS24143- (plus dash number from table V).

General characteristics: See table V.

Qualification by similarity: See table VI.

TABLE V. Dash numbers and general characteristics. 1/

| Dash number MS24143- | Type | Coil type | Terminal type | Mounting or mating socket | Auxiliary contacts | Maximum weight in pounds 2/ |
|---|---|---|---|---|---|---|
| D1 | I | dc | Stud | Flange | None | 1.5 |
| D2 | | dc | | | Yes | 1.7 |
| D3 3/ | | dc | | | Yes | 1.7 |
| A3 | | ac | | | None | 1.7 |
| A4 | | ac | | | Yes | 1.9 |

1/ A1 and A2 have been superseded by A3 and A4.
2/ Weights include covers and barriers.
3/ For Government logistics support, MS24143-D3 shall be used in lieu of MS21330-D2.

TABLE VI. Qualification by similarity.

| Dash number MS24143- | Loads | | | | | | Dynamics 1/ | | | Environmental 2/ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Type 1 | | | Type 1 ER | | | | | | | | |
| | A | B | C | D | E | F | A | X | Y | A | B | C |
| D1 | 4 | | | | | | 2 | | | 4 | | |
| D2 | 4 | | | | | | 2 | | | 4 2/ | | |
| D3 3/ | 4 | | | | | | 2 | | | 4 2/ | | |
| A3 | | | 1 | | | | | | 1 | | | 1 |
| A4 | 2/ | | 1 2/ | | | | | | 1 | | | 1 2/ |

1/ All relays must be tested. Reference MIL-R-6106, appendix I.
2/ Testing of relays with auxiliary contacts: Reference MIL-R-6106, appendix I.
3/ MS24143-D3 shall receive complete group C testing for retention of qualification.



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CONCLUDING MATERIAL

Custodians:
Air Force - 85
Navy - AS

Review activities:
Air Force - 99
Navy - EC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0921)



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